

Interferometric Study of Pinch Phase in Plasma-Focus Discharge at the Time of Neutron Production

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Abstract—A plasma column generated in the PF-1000 device working in deuterium gas at a current level of 1 MA was investigated with interferometric diagnostics and scintillation detectors. The beam of diagnostic laser of 527-nm wavelength was optically split into 16 beams with a time delay in the range from 0 to 220 ns. This diagnostic tool makes possible the imaging of the evolution of pinch geometry, the axial and radial distributions of plasma density in the column at the stagnation phase, and their comparison with the evolution of X-ray and neutron production. The evolution of dense structure is described with respect to its importance for fusion processes.

Index Terms—Neutron production, plasma diagnostics, plasma focus (PF).

I. INTRODUCTION

THE FAME of plasma focus (PF) has been based on the fact that it was the very intense neutron-producing device [1]–[4]. The scaling laws for the neutron yields formulated at the beginning of the PF investigations were very promising for these devices. However, later investigations carried out on bigger devices suggested that there is a certain condenser energy limit above which the scaling law is not valid. Hence, the essential problem to be resolved in PF research has always been to discover a mechanism of deuteron acceleration and neutron production.

The PF-1000 facility at the IPPLM Warsaw has been used for neutron diagnostics. The neutrons are produced mainly in a spherical region of 2 cm in diameter located at 6–10 cm in front of the central electrode [5]. From the neutron energy distribution, it follows that the fast deuterons in the energy

range of tens and hundreds of kiloelectronvolts move dominantly downstream [6], [7].

The plasma density of the densest part of the pinch column above 10^{25} m^{-3} was measured with a laser interferometry technique in different PF devices [8], [9].

Now, the 16-frame Mach–Zehnder interferometer has been established on the PF-1000. The interferometer is able to register pinch evolution during 220 ns in one shot. Thus, we have an opportunity to correlate neutron emission with pinch plasma dynamics (together with good time resolution).

In this paper, we concentrate on the description of the plasma evolution during the stagnation phase at a lower current amplitude of 1 MA and at a small total neutron yield of about 10^9 . In Section II, we describe an experimental setup and diagnostics. In Section III, we present the evolution of plasma column and the internal distribution of linear and plasma densities, and we compare them with neutron signals. The results are summarized in Section IV.

II. EXPERIMENTAL SETUP AND DIAGNOSTICS

Measurements were performed on the PF-1000 facility, operated at a lower energy of 120 kJ, a charging voltage of 20 kV, a maximum current that is equal to about 1.2 MA, and an initial deuterium pressure of 200 Pa (in comparison with 30 kV and a pressure of 400 Pa at the standard energy level). These conditions produced weaker plasma compression. Then, the electron density in the plasma column below 10^{19} cm^{-3} and the slower transformation of the plasma column enabled a better distinct path and a contrast of fringe patterns. These conditions mentioned earlier were not fulfilled at the higher operating energy. The plasma column formed in the discharges at the lower energy has higher diameter, worse cylindrical symmetry, and lower stability of the stagnation. The facility was equipped with the same Mather-type coaxial electrodes as in shots with higher energy. The length of both electrodes was 480 mm. The anode has a diameter of 230 mm. The outer cathode of 400 mm in diameter consisted of 12 stainless-steel rods, distributed symmetrically.

X-rays were measured with silicon PIN detectors filtered with a Be foil of 10- μm thickness. Four BC-408 plastic scintillation probes of 5-cm thickness coupled with fast photomultipliers were used to perform the detailed time-resolved measurements of neutrons and hard X-rays (HXR) with energy above a few hundreds of kiloelectronvolts (a penetrating

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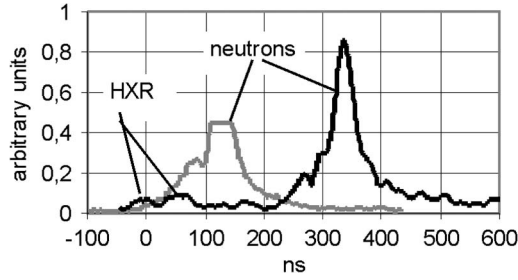


Fig. 1. Signals from scintillation detectors registered (gray) side-on at a distance of 1.7 m and (black) downstream at a distance of 7 m.

chamber steel wall of 2–3-cm thickness). The probes were situated downstream and upstream at distances of 7.0 m and side-on at distances of 1.6 and 7.0 m. For optical diagnostics, the Nd:YLF laser working at the second harmonics was used. One laser pulse (527 nm; 2-ns duration) was split with mirrors in the unique system of 16 beams. These beams pass the Mach–Zehnder interferometer configuration and the plasma with time delays of 10, 30, 40, 60, 70, 90, 100, 120, 130, 150, 160, 180, 190, 210, and 220 ns after the first beam. All interferograms were registered on photographic films.

III. EXPERIMENTAL RESULTS AND DISCUSSION

We describe shot number 8313, in which the exposition of interferograms covered the stagnation phase of the pinch at the time of neutron production with a total neutron yield of 3×10^9 . The peak of the PIN detector signal was assigned to the zero time.

The registered signals of neutrons are influenced with scattered and moderated neutrons. These neutrons have lower velocity and energy in comparison with the neutrons going straight from the pinch to detector, and they are registered mainly in the tail of the signal. Moreover, the neutrons produced in the cold gas in the chamber out of the discharge volume reached the detector with a time delay. Fast deuterons pass the distance above 10–20 cm during 50–100 ns. The number of delayed neutrons is roughly 50%. We assume that the start and maximum of signals are not influenced by the delayed neutrons.

The signals of HXRs and neutrons registered with scintillation detectors are shown in Fig. 1. The scintillation detector placed at a 7-m distance from the center of the pinch column downstream (black) registered both HXRs and neutrons. The side-on detector at a distance of 1.6 m was filtered with 4 mm of lead (to absorb HXRs). HXRs and neutrons were emitted in two pulses. The first HXR pulse started at a time of -30 ns, reached maximum at 0 ns, and finished at 30 ns. The second HXRs started at 40 ns, their maximum was at 70 ns, and they finished at 100 ns. Neutron signals were processed by the time-of-flight method. The first neutron pulse was much smaller than the second. Both started practically at the same time as HXRs, and their FWHM was 50 ns. A dominant portion (70%) of neutrons in the first peak was produced during 50 ns. The maximum of neutron signal registered at 7 m downstream was 30 ns before the upstream one. Then, the dominant neutron

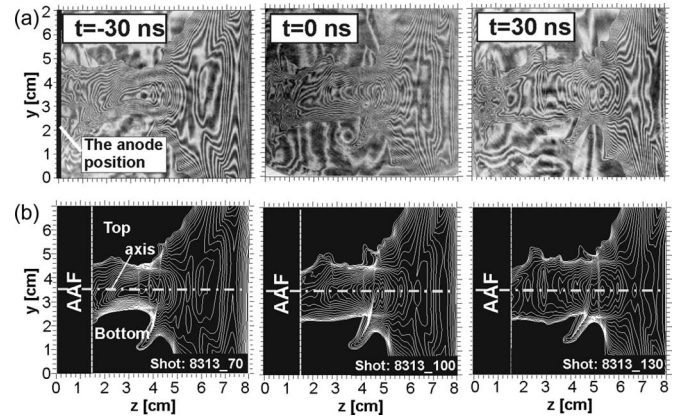


Fig. 2. Shot 8313. Interferograms registered at -30 , 0 , and 30 ns: (a) Original interferograms and (b) their reconstructions obtained by the automated method of the maximum fringes. (AAF) Area of the string asymmetry of interferometric fringes.

energy downstream was 2.7 MeV, and the dominant energy upstream was 2.25 MeV. Therefore, deuteron-producing neutrons moved mostly downstream with an axial energy component of about 50 keV. These characteristics determined the dominant beam–target mechanism of neutron production.

The interferograms were registered at -100 , -90 , -70 , -60 , -40 , -30 , -10 , 0 , 20 , 30 , 50 , 60 , 80 , 90 , 110 , and 120 ns. The temporal uncertainty between interferograms and neutron signals was below 5 ns.

In Fig. 2, three typical interferograms are shown. The top sequence in Fig. 2(a) shows the original interferograms of the plasma column, while their fringe pattern reconstructions are shown in Fig. 2(b) (the bottom sequence).

The reconstructed interferograms were used for the calculation of the electron density distributions in the plasma column on the basis of the methodology described in [10]. The first step of this methodology consists in the determination of the phase distribution by the automated method of the maximum fringe determination. Such way of phase distribution is necessary to use for the high shifting of the interferometric fringes and the complicity of their structure. Due to the asymmetry of the PF column, the electron density distributions were calculated for the top half of the interferogram [above the symmetric axis (see the Fig. 2)].

Then, the electron density distributions were calculated by means of the fast Fourier transformation [11]. The wings at the boundary have no essential influence both on the qualitative and quantitative characters of the linear density profiles and on the total electron number. It is particularly important because these distributions are the basic source of information about the fine structures of about 1-mm dimensions. In addition, for the interpretation of the experimental results, the area (AAF) near the anode (for $z < 1.5$ cm) was not taken into account due to a strong asymmetry of the interferometric fringes [see Fig. 2(a)].

The electron densities calculated from the interferograms in Fig. 2(b) are shown in Fig. 3. To characterize the formation and evolution of the stagnation plasma column on the PF-1000 device, the linear electron density distributions have been calculated from the electron density distributions. The results of the calculation are shown in Fig. 4.

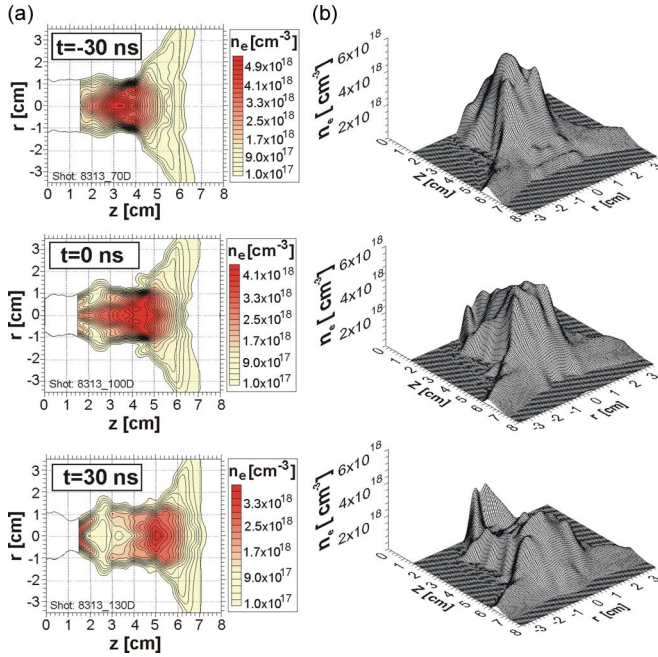


Fig. 3. Equidensities of the interferograms at -30 , 0 , and 30 ns shown in Fig. 1.

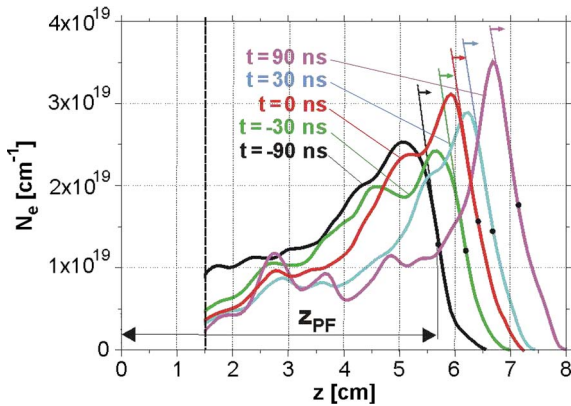


Fig. 4. Linear electron density distributions at different times of the PF column dynamics.

The sequence of the linear electron densities demonstrates the evolution of the plasma column and its front during a -90 – 90 -ns period. To calculate the plasma front velocity, we estimate the position of the front z_{PF} from the half maximum of the linear electron density (see Fig. 4). The results are shown in Fig. 5. The plasma column front moves with a velocity (V_{PF}) of below 10^7 cm/s.

The electron density distributions shown in Fig. 3 show that the formed plasma column is not homogeneous along the axial and radial axes. The electron density has the local maximum, which is clearly visible in Figs. 3 and 4. The axial electron density distribution (obtained at $r = 0$) for different times is shown in Fig. 6. This diagram shows that the maximum of the electron density appears near the anode and achieves a value of almost $6 \times 10^{18} \text{ cm}^{-3}$.

The aforementioned results clearly demonstrate the existence of a dense plasma structure, which moves along the axis from the anode. Its density decreases, and its FWHM increases with

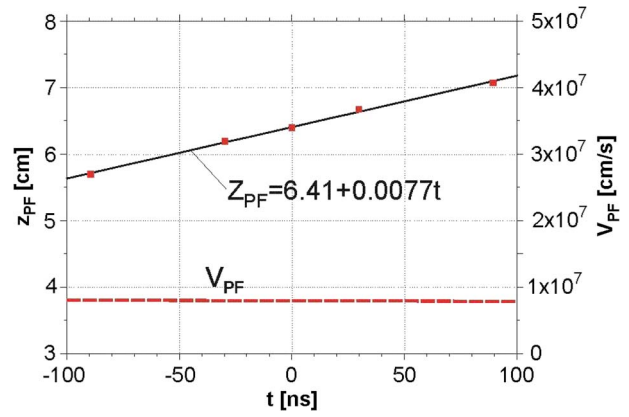


Fig. 5. Temporal dependence of the plasma column front (z_{PF}) position and its velocity (V_{PF}).

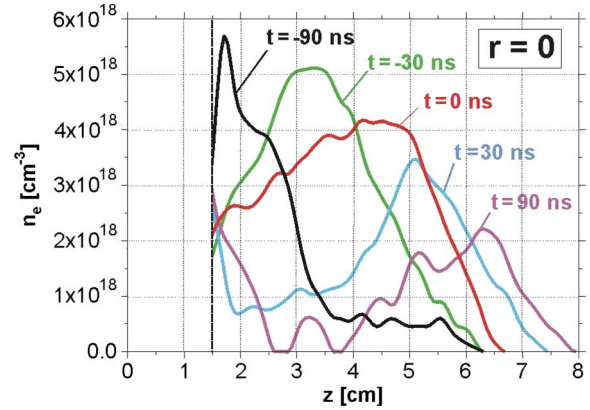


Fig. 6. Temporal dependence of the axial of electron density in the plasma column.

time. This structure correlates spatially with the source of fusion protons [5].

In order to obtain more details about the parameters and the dynamics of this plasma structure, the linear electron density distributions and the total electron densities in the plasma column were calculated for the limited plasma radius (r) in the range from -1.5 to 1.5 cm only. Such treatment allows us to separate the central configurations from the external part of the plasma (see the irregular lobes and umbrellalike shape of the plasma front in Fig. 2). Then, we can determine the velocity of the front of these configurations, as well as the total number of electrons corresponding to different times. The linear electron density distributions and the electron densities, which were taken into calculations, are presented on the upper part of the images in Fig. 7. The evolution of the linear electron densities is shown in Fig. 8. In Fig. 8, we can see the same angle α of the slope of the linear electron density distributions from the z -axis (see Fig. 7). It is constant during the observed period $-70 \div 90$ ns.

The velocity of the plasma current sheath front was determined on the basis of the distributions of the linear electron density. Analogically, as in the case of the total region of the plasma column, the velocity of the plasma front was determined on the basis of the position of the half of maximum at the signal onset z_{PS} . The temporal dependences of the plasma

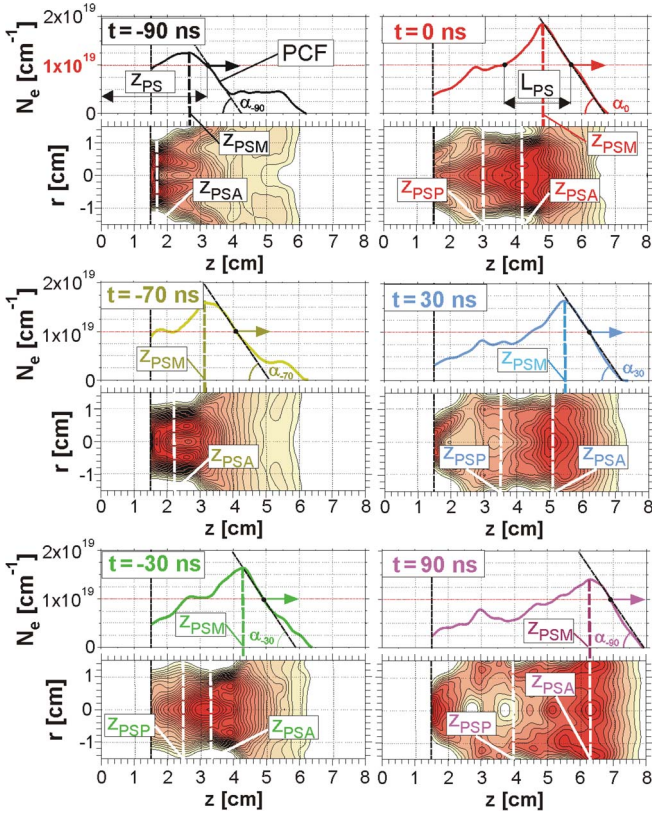


Fig. 7. Linear electron density distribution and the linear density in the plasma illustrating the movement of the plasma configuration. (z_{PSA}) Maximum of electron density. (z_{PSM}) Maximum of linear density. (L_{PS}) Dimension of the plasma structure.

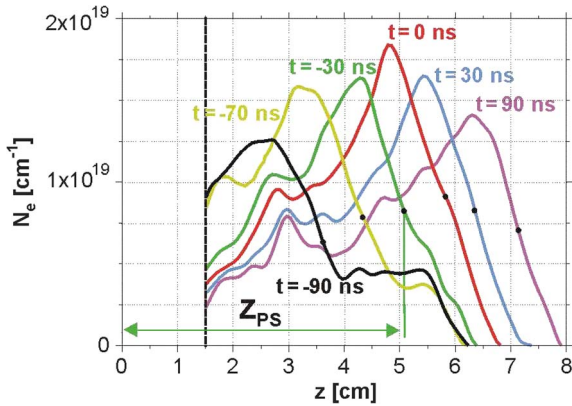


Fig. 8. Temporal evolution of the linear electron density distributions in the plasma column.

configuration front position and the velocity (V_{PS}) of this front are shown in Fig. 9. The diagram shows that the velocity of the moving dense structure of plasma decreases linearly from a value of 3.2×10^7 cm/s (at -100 ns, i.e., during the implosion) down to a value below 1×10^7 cm/s (at 100 ns, i.e., during the expansion).

The linear density profiles enable us to estimate the internal evolution of the dense plasma structure, which has a dimension of 2–3 cm.

The plasma column structure in perpendicular direction can be characterized by the radial electron density profiles. The

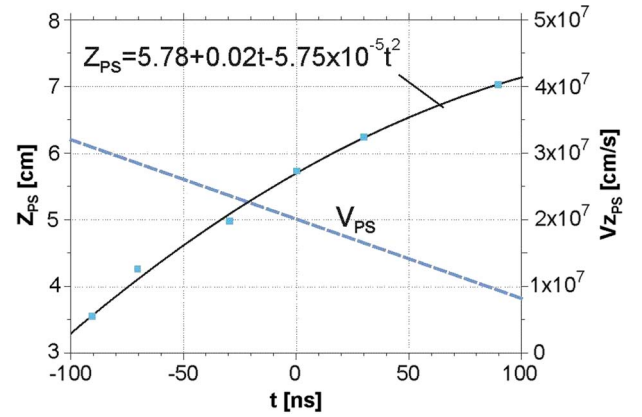


Fig. 9. Temporal dependence of plasma current sheath front position (z_{PS}) and the calculated velocity (v_{PS}).

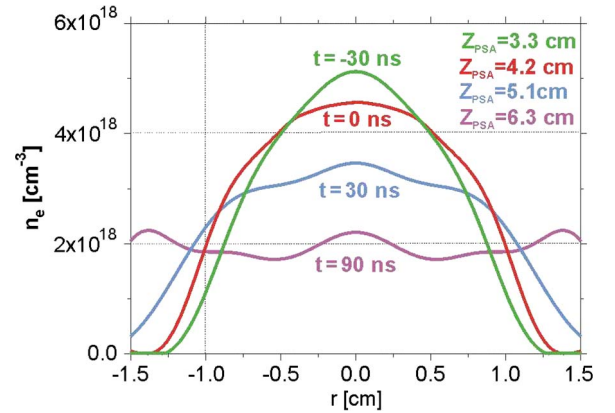


Fig. 10. Radial electron density profiles for the peak of the dense plasma structure (z_{PCA}) at different times.

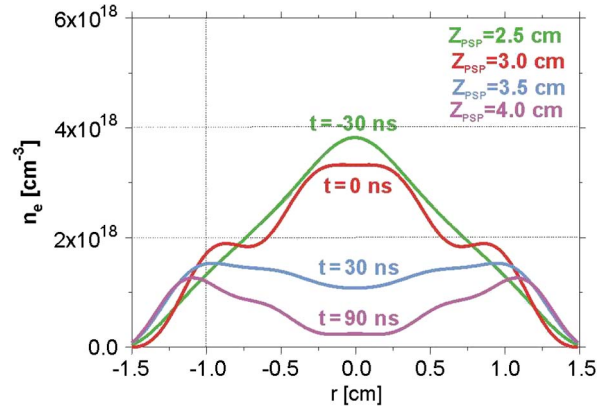


Fig. 11. Radial electron density profiles in the plasma region with low density in z distance in Fig. 6 at different times.

radial electron density profiles at the peak of density in Fig. 6 are shown in Fig. 10, while Fig. 11 shows the radial electron density profiles corresponding to the density minimum in Fig. 7 at the times of neutron production. From the comparison of the density distributions shown in Figs. 10 and 11, we can see differences between the dense plasma structure and the plasma with a low electron density in the column. In the case of the dense plasma structure in Fig. 10, the radial profiles of

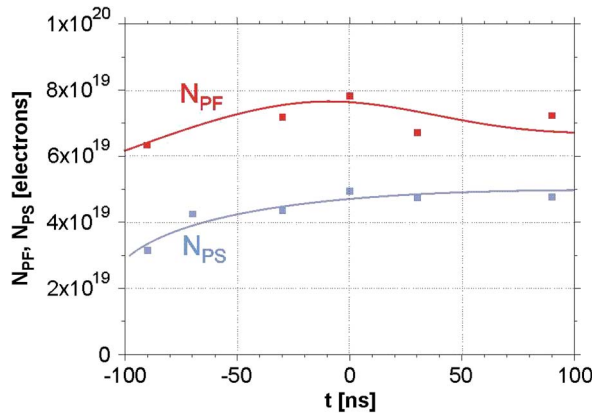


Fig. 12. Temporal dependence of the total electron number: For the whole observed column N_{PF} and the dense structure N_{PS} .

densities have characteristic maxima on the symmetry axis, and these maximum values are higher than the electron densities at the boundaries. The electron densities in the regions with a low density are considerably smaller in comparison to the dense plasma structure, and additionally, these profiles have a minimum on the axis clearly visible in the time of high HXR and neutron production from $t = 30$ to 90 ns (Fig. 11). For $t = 90$ ns, the electron density in the axis region is more than one order smaller in comparison with the dense plasma structure (Figs. 10 and 11).

The obtained electron density distributions allow us to calculate the number of electrons in the whole column and in the dense structure. The temporal dependence of the total electron number in the case of the whole observed column of the plasma (N_{PF}), as well as the dense plasma structure (N_{PS}), is shown in Fig. 12.

Because the investigated deuterium plasma in the PF-1000 device is fully ionized, the presented electron distributions correspond to the deuteron distributions. The curves in Fig. 12 show the total number of deuterons as a function of time. The number of deuterons in the total column is about 7×10^{19} . This number in a dense structure of $4 \div 5 \times 10^{19}$ is roughly constant during the neutron production. The dense structure is a region with dominant D–D fusion collisions. The plasma density $n_e = 4 \times 10^{18} \text{ cm}^{-3}$, the dimension of the target $l = 2 \text{ cm}$, the cross section of D–D reaction $\sigma(50 \text{ keV}) = 4.6 \times 10^{-27} \text{ cm}^2$, and the total neutron yield $NY = 3 \times 10^9$ enabled us to estimate the number of accelerated fast deuterons N_D [7]

$$N_D = 2 \frac{NY}{\sigma(50 \text{ keV})n_e l} \approx 10^{17}.$$

A factor of two represents the two branches of D–D reaction. These deuterons escape the pinch practically without collisions. The number and energy of fast deuterons make possible the estimation of their total energy of 2.6 kJ and current of 35 kA.

The low-density region seems to be the locality of acceleration of deuterons. The total number of deuterons escaping the locality in the axial range from $z = 3$ to 4.5 cm during 30–90 ns amounted to 10^{18} (Figs. 4 and 8). The radial and axial velocities observed at the moving plasma (roughly 10^7 cm/s) were estimated from temporal dependences imaged in the

figures. They enabled the evaluation of the mean energy of the dominant part of the transported deuterons in the range of a few kiloelectronvolts. This energy is insufficient for fusion collisions. The velocities of deuterons producing neutrons must be one order higher, and these velocities in the registered interferograms were not visible.

In interferometry diagnostics, we have no images of filamentary structure mentioned in [1] and of dense spots with submillimeter dimensions. The variation of density has a dimension on the order of centimeters.

IV. SUMMARY AND CONCLUSION

The results obtained with the processing of neutron signals and interferograms in one shot number 8313 with a lower current of 1 MA and a neutron yield of 3×10^9 at the PF 1000 device can be summarized as follows.

HXRs and neutrons were emitted in two pulses with an FWHM of 30 ns. The second one was more intensive with a downstream dominant energy of about 2.7 MeV. The beam–target mechanism of neutron generation was caused by deuterons moving downstream with an energy of about 50 keV.

The plasma density during stagnation has not been uniform both in the radial and axial directions. At the implosion of the current sheath, a dense structure with a radius of 2 cm is formed. Its maximum density at the center has reached $2\text{--}4 \times 10^{18} \text{ cm}^{-3}$ and decreased with time. The number of deuterons in a dense structure of $\sim 5 \times 10^{19}$ has been constant during the neutron production. This dense structure is the target with fast deuterons, and thus, it is the source of neutrons. The beam of deuterons carries a current of $\sim 35 \text{ kA}$, and it contains $\sim 10^{17}$ particles with a total energy of $\sim 1 \text{ kJ}$.

A locality with a small plasma density has been formed at the central part between the anode and the dense structure after the time 0 ns. It is probably a locality of deuteron acceleration.

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